



Artificial Intelligence-Driven Transformation in Financial Technology: Applications, Agents and Challenges

Zhizhou Zhang^{1,2,*} and Meiqi Lu^{3,*}

Abstract

Artificial intelligence (AI) has become a pivotal force in the financial technology (fintech) sector, reshaping services through enhanced automation, improved efficiency, and personalization. The integration of sophisticated AI models, however, introduces significant challenges concerning data privacy, algorithmic bias, and a lack of transparency, which necessitates a systematic and comprehensive evaluation of the field. This review provides a holistic synthesis of the current landscape, introducing a structured taxonomy of AI applications that ranges from foundational machine learning in areas like credit scoring and fraud detection to advanced autonomous agents capable of dynamic decision making. It systematically analyzes critical technical and ethical hurdles, including model interpretability, data quality issues, and security vulnerabilities. Furthermore, the review examines emerging paradigms, such as the deployment of autonomous and multi-agent systems that are revolutionizing financial workflows and strategies. By bridging the gap between AI and financial applications and identifying unresolved challenges, this analysis delineates a clear agenda for future research aimed at fostering a more robust, transparent, and equitable AI-driven financial ecosystem.

Keywords: Artificial intelligence; Financial research; Machine learning; Artificial intelligence agent; Financial technology.

Received: 05 February 2026; Revised: 29 March 2026; Accepted: 04 May 2026

Article type: Review article.

1. Introduction

Artificial intelligence became increasingly important to financial technology because financial systems generated large scale, high frequency, and highly heterogeneous data that exceeded the analytical capacity of many conventional decision frameworks.^[1,2] The digitalization of payments, lending, investment, insurance, and platform based services further accelerated this transformation, since financial institutions and fintech firms needed methods that could process transactional records, behavioural signals, text, and alternative data in near real time.^[3,4] Prior studies showed that this shift was not only technical but also structural, because AI supported new forms of service delivery, customer interaction,

risk monitoring, and product personalization across the financial sector.^[5] The growing relevance of AI in fintech was therefore tied to both economic value creation and institutional adaptation, as firms sought greater operational efficiency, faster decisions, and broader market reach in increasingly data intensive environments. More recently, this background evolved beyond predictive analytics alone, as finance began to incorporate AI agents capable of interacting with data, tools, and decision environments in more adaptive and semi-autonomous ways.^[6,7] Emerging studies suggested that these agent based systems could extend the role of AI from classification and forecasting toward sequential decision support, regulatory reasoning, portfolio adjustment, and task orchestration across financial workflows.^[8,9]

The importance of AI in fintech was also reinforced by the limitations of conventional rule based and linear modelling approaches in settings marked by uncertainty, fraud risk, market volatility, and rapidly changing customer behaviour.^[10–12] In lending, machine learning methods improved the ability to identify borrower risk and extract information from complex or nontraditional variables, while in fraud detection

¹School of Engineering, The University of Manchester, Manchester, England, M13 9PL, UK

²Faculty of Engineering, University College London, London, England, WC1E 6BT, UK

³The Bartlett Faculty of the Built Environment, University College London, London, England, WC1E 6BT, UK

*Email: zhizhou.zhang@manchester.ac.uk (Z. Zhang),

meiqi.lu.22@ucl.ac.uk (M. Lu)

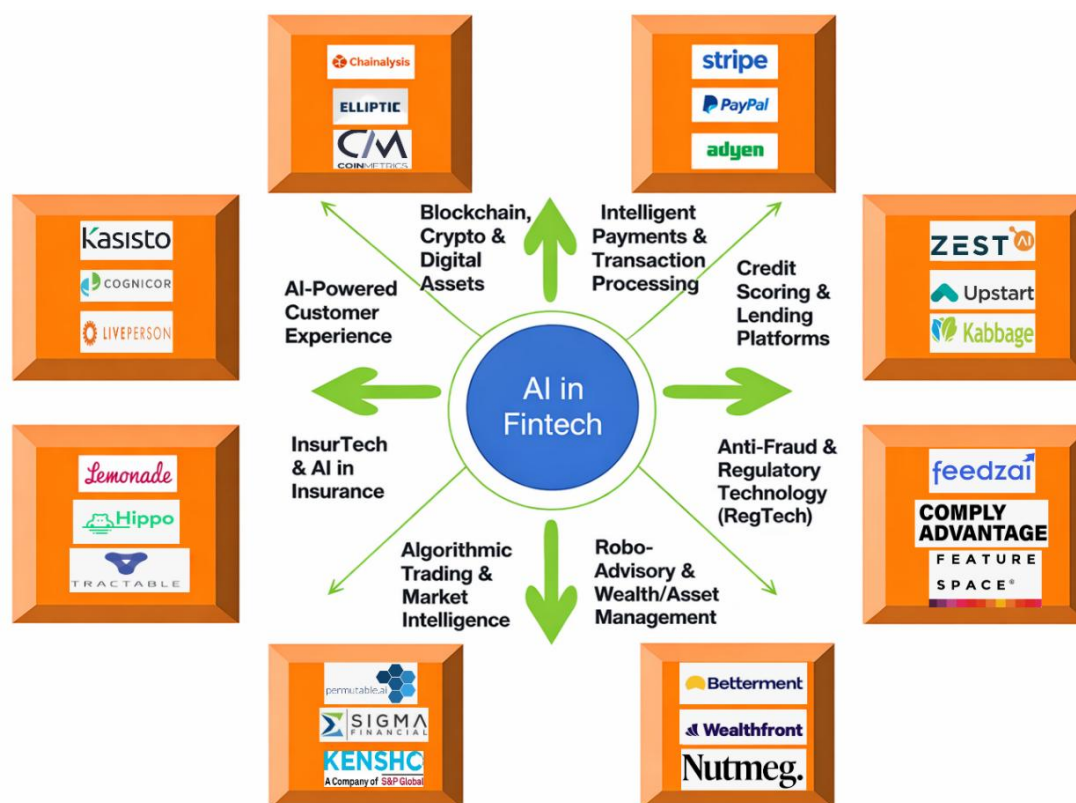


Fig. 1: A market map of key artificial intelligence (AI) application areas in the fintech industry.

and transaction monitoring, AI based systems strengthened anomaly recognition and adaptive surveillance capacities.^[13–17] In financial markets, deep learning, reinforcement learning, and multi agent architectures were increasingly used for forecasting, sentiment extraction, trading support, and dynamic portfolio management, which reflected the broader movement of finance toward automated and data driven decision environments.^[18–20] At the customer interface, AI also became important through chatbots, advisory tools, and intelligent service systems that reshaped communication between financial institutions and users.^[21–23] At the same time, the rise of AI agents made the background of fintech more consequential because these systems raised stronger questions about interpretability, accountability, compliance, and human oversight in high stakes financial settings.^[24–26] AI became important in fintech not simply because it improved prediction, but because it began to reshape how financial value was assessed, how decisions were made, and how increasingly autonomous financial services were delivered and governed across the digital economy.^[27–30]

Artificial intelligence (AI) has emerged as a transformative force in the financial technology (fintech) sector, fundamentally reshaping how financial services are delivered, managed, and consumed.^[31] Fig. 1 provides a schematic overview of the primary application domains of artificial intelligence within the fintech sector, showcasing

representative companies operating in each key area such as intelligent payments, credit scoring, anti-fraud technology, and algorithmic trading. The integration of AI is driving a paradigm shift from traditional financial models toward more automated, intelligent, and personalized systems. This evolution, fuelled by breakthroughs in machine learning, natural language processing, and big data analytics, has positioned AI as a cornerstone of innovation within the financial industry.^[13,32] Financial institutions are increasingly leveraging AI to enhance operational efficiency, improve risk management and decision making processes, and revolutionize the customer experience.^[33,34] The impact of AI now spans a wide array of financial applications, including algorithmic trading, fraud detection, credit scoring, and personalized financial advisory services.^[35–37]

The AI in fintech market is experiencing substantial growth, with projections indicating its global size will expand significantly in the coming years. Generative AI market report^[38] illustrated that the global generative AI market is projected to expand from USD 13.5 billion in 2023 to approximately USD 255.8 billion by 2033, representing a compound annual growth rate (CAGR) of 34.2% over the forecast period. In 2023, North America established market primacy, accounting for over 42.1% of global revenue, valued at USD 5.6 billion. This expansion is driven by the profound impact AI has on the financial industry, where it is no longer

a peripheral technology but a core component of business strategy and operations.^[39] The importance of AI in this sector stems from its ability to process and analyze massive datasets, leading to enhanced efficiency, improved accuracy in decision-making, and the creation of innovative, personalized financial services.^[40] Financial institutions are increasingly leveraging AI to automate routine processes, which minimizes operational costs and allows human resources to focus on more complex tasks.

Prior work^[11,13,14] showed that the state of the art in artificial intelligence for fintech first developed around prediction intensive tasks, particularly credit assessment, lending analytics, and fraud detection, where machine learning methods were increasingly adopted to outperform many conventional statistical and rule based approaches under complex data conditions. The historical application of AI in finance began with early computational models and rule based expert systems for tasks like algorithmic trading^[41] and fraud detection.^[42] However, the recent explosion in data availability and computational power has enabled the deployment of more sophisticated AI models, such as deep neural networks, which can uncover complex, non linear patterns in vast datasets. This has led to significant improvements in predictive accuracy for tasks like market trend forecasting^[43] and credit risk assessment.^[33] The rise of generative AI has further expanded the capabilities of financial institutions, allowing for the automation of complex knowledge work such as financial report generation,^[44] sentiment analysis,^[45] and the creation of synthetic data^[46] for robust model training. As a result, AI is no longer a peripheral technology but a central driver of competitive advantage and innovation in the global financial ecosystem. Fintech lending studies reported that alternative data and machine learning improved borrower screening and expanded the informational basis of credit evaluation, especially for applicants who were less well served by traditional credit infrastructures.^[13] Benchmarking studies in credit scoring further showed that ensemble learning and other advanced classifiers often delivered stronger predictive performance than many earlier baseline models, which made AI driven credit modelling one of the most mature areas in financial technology research.^[14] Explainable machine learning in credit risk management became an important strand of research in response to regulatory scrutiny and the need to justify automated financial decisions.^[33] Recent systematic reviews^[24,25] confirmed that explainable artificial intelligence had become a major component of finance research because transparency, accountability, and traceability were increasingly treated as core requirements in high consequence financial settings. In parallel, fraud detection and

anti money laundering studies established another major state of the art stream, showing that AI supported adaptive anomaly detection, transaction surveillance, and operational monitoring across payment systems and financial crime control.^[12,17] The technical frontier had been shaped by large scale data engineering, hybrid modelling, and the need to respond to continuously evolving fraud tactics rather than static risk patterns.

The state of the art then expanded from risk screening toward market intelligence, financial decision support, and customer facing service systems. In financial markets, prior work showed that deep learning, time series forecasting, sentiment analysis, and reinforcement learning were increasingly used for trading, portfolio selection, and market prediction, reflecting a broader shift from static estimation toward dynamic and data responsive finance.^[18–20] Research on AI powered chatbots and advisory systems also indicated that fintech applications had moved into customer service, compliance support, segmentation, and personalized financial interaction, which demonstrated that AI was no longer confined to back end risk modelling alone.^[21,23] The frontier of the field had entered a new phase characterized by explainability, generative AI, and finance specific large language models, because financial institutions increasingly needed systems that could interpret text, support regulatory reasoning, and assist with knowledge intensive financial tasks.^[47] Emerging work on intelligent financial systems indicated that AI in finance was developing from narrowly defined predictive tools toward more integrated decision infrastructures that combined data analysis, language processing, and adaptive support across multiple financial functions. The contemporary state of the art in fintech had been defined not only by stronger predictive models, but also by the growing integration of explainability, language based intelligence, and increasingly agent like forms of financial decision support.

Despite the rapid diffusion of artificial intelligence in financial technology, its practical deployment remained constrained by severe data related limitations. Financial datasets were often fragmented across institutions, restricted by privacy regulation, and affected by noise, missing values, reporting inconsistency, and regime shifts, all of which reduced model portability and weakened out of sample reliability.^[4,47] Privacy preserving collaboration therefore became an important research direction, yet decentralized and federated learning approaches still introduced additional communication cost, computational burden, and security exposure, and they did not fully resolve the tension between privacy protection and predictive utility.^[14,25,33]

A second major gap concerned transparency and fairness in high stakes financial decisions. Many high performing models in lending, fraud detection, and risk monitoring still operated as black boxes, while recent reviews in finance showed that explainability methods had not yet been consistently validated for domain specific decision support, regulatory audit, or stable local interpretation across settings.^[24] Fairness remained difficult to define and operationalize in credit contexts because different fairness criteria could conflict with each other and with profitability objectives.^[12] Recent evidence from credit rating research further showed that fairness and explainability were tightly connected, yet joint evaluation frameworks remained limited and unfairness could still emerge in smaller or subgroup specific samples.^[17] Robustness under adversarial and dynamic conditions also remained insufficiently addressed. Research across industrial machine learning showed that adversarial manipulation had become a realistic concern, while defensive practice remained uneven and often immature at the deployment stage.^[18] In fintech, this challenge became more acute as transformer based credit scoring models that used borrower text were shown to be vulnerable to small semantically neutral perturbations, which could materially alter model outputs and create risks of gaming, instability, and unreliable decision making.^[19]

Another emerging gap concerned the use of large language models and generative systems in finance. Recent research showed that hallucination remained a central weakness of large language models, particularly in specialized domains that required factual precision and verifiable reasoning.^[48] Retrieval augmented generation was increasingly proposed as a mitigation strategy because it improved grounding through external knowledge access, yet current evidence also showed that retrieval support did not eliminate harmful or unreliable outputs and that model performance still varied substantially across tasks and contexts.^[49,50] Although the existing literature^[4,48,51] had provided valuable insights into specific aspects of artificial intelligence in fintech, most reviews had treated machine learning applications, explainable artificial intelligence, and regulatory governance as separate lines of inquiry. A comprehensive synthesis that brought together conventional predictive models, generative systems, and emerging agent based architectures within a unified fintech framework, while simultaneously addressing governance, security, fairness, and financial stability, had remained relatively limited.

This review provided a holistic synthesis of the current state of artificial intelligence within the financial technology sector. It introduced a structured taxonomy that classified

applications across key financial domains and mapped them to underlying models, ranging from foundational machine learning algorithms to advanced generative artificial intelligence and autonomous agents. A primary contribution of this work involved a detailed examination of emerging paradigms, with a particular focus on the rise autonomous agents. The research analyzed agent architecture and the potential for these systems to revolutionize financial workflows through complex reasoning. Furthermore, the analysis documented the operational shift from task specific automation toward holistic and intelligent financial ecosystems. The review also provided a systematic analysis critical technical and ethical challenges that hindered the responsible adoption of artificial intelligence in finance. It assessed persistent issues including data privacy, model interpretability, algorithmic bias, security against adversarial attacks, and the complexities of regulatory compliance. Finally, by identifying unresolved challenges and exploring the convergence regarding artificial intelligence with other frontier technologies, the study delineated a clear agenda for future research aimed at developing more robust, transparent, and equitable financial systems.

2. AI-driven transformation in fintech

Artificial intelligence (AI) has rapidly emerged as a cornerstone of modern financial technology, reshaping how institutions and consumers interact with financial systems.^[31] By leveraging machine learning, natural language processing, and deep learning, AI-driven fintech extends far beyond traditional automation, enabling predictive analytics, personalized financial services, and real-time risk management. Recent advances in large language models and generative AI agents further accelerate this transformation by enhancing interpretability, adaptability, and decision-making across lending, payments, trading, compliance, and customer engagement.^[52] As a result, AI is no longer a peripheral tool but a central enabler of innovation, efficiency, and competitiveness in the global financial ecosystem. This chapter explores the core applications, benefits, and challenges of AI integration into fintech, highlighting both the technological breakthroughs and the governance frameworks necessary to ensure fairness, accountability, and trust in these high-stakes environments.

2.1 Reasons to use machine learning

Artificial intelligence offered clear advantages for the modern financial technology stack because it processed large and heterogeneous data at scale, automated complex decisions, and supported end to end digital services across lending,

payments, wealth, and compliance. Survey work in smart financial technology and in banking documented how artificial intelligence systems increased decision quality, supported full digital workflows, and reduced operational frictions across risk, service, and analytics functions.^[1,2]

Credit risk management benefited most visibly. Lender studies showed that machine learning models that used alternative data improved default prediction and widened access for thin file borrowers while preserving or improving risk grading. Evidence from a fintech platform demonstrated that rating grades that incorporated nontraditional data predicted performance and reclassified some borrowers into better priced loan grades.^[13] Comparative experiments with phone metadata, psychometrics, and other sources reported accuracy gains over traditional demographic only baselines.^[15] Classic reviews in credit risk established the methodological foundation for these developments and explained why non linear models and richer features improved credit scoring in practice.^[10]

Artificial intelligence also enhanced market sensing and trading. Deep reinforcement learning studies framed portfolio rebalancing as a sequential decision problem and reported improved risk adjusted metrics in backtests when reward functions targeted portfolio objectives directly.^[18] At the same time, progress in deep learning for time series forecasting expanded the toolkit for multi horizon prediction that underpinned risk, pricing, and allocation engines in financial applications.^[19] Text analysis of financial news further showed that quantified media tone and salience affected price formation, which motivated the use of natural language models in trading and surveillance pipelines.^[20]

Fraud and financial crime programs gained measurable efficiency from artificial intelligence. Cost sensitive and champion challenger set ups in card authorization streams improved savings by raising detection with fewer false positives relative to static rules, which translated into lower manual review volumes and faster time to decision.^[16] In anti money laundering, qualitative fieldwork with banks and providers highlighted pain points in transaction monitoring and pointed to machine learning as a route to reduce investigative backlogs and false alerts while preserving coverage of evolving typologies.^[53] A methods review from the information systems perspective catalogued the data engineering and model governance practices that made fraud analytics reliable at scale.^[17]

Customer engagement and service operations also benefited. Institutions deployed conversational agents to deliver always on support for balance queries, payments, and account changes, which reduced wait times and agent

workload. A broad review of chatbot technologies described how advances in natural language processing and dialog management supported these service gains in financial settings.^[22] Experimental evidence in electronic markets research further reported that artificial intelligence chatbots shaped user behavior and compliance in service flows, indicating value for onboarding, education, and guidance use cases.^[21]

Finally, the adoption of explainable artificial intelligence strengthened trust, audit, and regulatory alignment. Recent work on credit scoring combined deep learning with post hoc explanation techniques and demonstrated that high performing models could be interrogated for driver analysis and policy review without sacrificing predictive power, which supported deployment under transparency requirements.^[36]

2.2 Open-source resources for AI in fintech

The rapid integration of artificial intelligence within the financial technology sector is substantially supported by the broad availability of open source tools and datasets. [Table 1](#) reviewed open-source resources that had been applied in finance and economics. These resources have democratized access to sophisticated analytical capabilities, enabling researchers and practitioners to develop, benchmark, and deploy complex models for financial applications.^[34] The ecosystem of open source software libraries provides the foundational building blocks for machine learning, while a growing number of specialized and general purpose datasets offer the empirical grounding needed for rigorous model training and validation.

A diverse range of open source datasets is crucial for advancing financial research. For complex tasks like numerical reasoning over financial documents, specialized datasets such as FinQA^[54] have been developed to train and evaluate models on question answer pairs derived from financial reports. To test a model's nuanced understanding of financial language, benchmarks like FinNLI^[55] offer a basis for assessing natural language inference from sources such as regulatory filings and earnings call transcripts. More comprehensive benchmarks, including FinBen,^[56] provide a holistic framework for evaluating large language models across a spectrum of financial tasks from information extraction to risk management. Beyond these specialized resources, extensive macroeconomic time series data are available from platforms like the Federal Reserve Economic Data (FRED) database,^[57] which is a cornerstone for economic forecasting and financial market research. General purpose repositories also serve the fintech community; platforms like Zenodo^[58] promote open science by hosting accessible

Table 1: Open-source packages, machine learning models, dataset and applications for fintech.

Machine learning	Description and features	Application
FinQA ^[54]	A large-scale dataset of 8,281 question-answer pairs over 2,800 financial reports, with numerical reasoning, combining structured and unstructured data (tables, text)	Numerical reasoning over financial documents; training models to answer finance questions; explainable AI; feature extraction for predictive models using financial statements.
FinNLI ^[55]	Benchmark for Natural Language Inference in financial text, using SEC filings / earnings call transcripts	Testing understanding of financial text entailment; fine-tuning NLP models for inference / summarization; detecting misleading statements or consistency checking
FinBen ^[56]	A holistic benchmark for financial LLMs, covering many datasets across financial tasks: IE (information extraction), QA, forecasting, text generation, decision, making and risk management.	Evaluating LLMs on finance tasks; benchmarking model performance across tasks; guiding architecture/hyperparameter choices; research into generalization for diverse financial tasks
Financial Reports SEC ^[68]	Contains US public firms' annual reports (10-K) from ~1993-2020.	Fine-tune LLMs on financial text; use for section extraction, sentiment analysis; training models to understand structure of 10-K filings.
The Federal Reserve Economic Data (FRED) database ^[57]	Offers extensive U.S. economic time series and indicators. ^[69]	Macroeconomic trend analysis, ^[70] Economic forecasting, ^[71] Financial market research, ^[72] Policy impact assessment. ^[73]
Google Dataset search ^[59]	Global dataset search, broad coverage, real-time updates. ^[74]	Finding datasets for machine learning projects, ^[75] Supporting academic research, ^[76] Sourcing data for data journalism, ^[77] Enabling reproducible scientific studies, ^[78] Market analysis and business intelligence, ^[79] Locating open government and health data
Open-source datasets		
NIST ^[80]	Standardized datasets, measurement data, scientific and technical benchmarks. ^[81]	Benchmarking machine learning models, ^[82] Testing algorithm performance, ^[83] Scientific research validation, ^[84] Developing measurement standards ^[85]
Zenodo ^[58]	Open-access research data, multidisciplinary, supports datasets and publications. ^[86]	Sharing open research datasets, ^[87] Assigning DOIs for easy citation, ^[86] Collaborative project data storage, ^[88] Hosting software and code archives
AmeriGEOSS Community Platform DataHub ^[89]	Environmental data sharing, geospatial datasets, supports Americas collaboration. ^[90]	Sharing environmental and geospatial data, ^[91] Supporting disaster management and response, ^[92] Facilitating cross-country scientific collaboration, ^[93] Monitoring climate and ecosystem changes, ^[94] Enabling open access to regional datasets, ^[95] Promoting sustainable development initiatives in the Americas
StockEmotions ^[60]	Stock market emotions, rich labels, real-time updates. ^[96]	Predicting stock price movement based on sentiment, ^[97] Analysing market reaction to news and events, ^[98] Training sentiment analysis models for finance, ^[99] Building trading strategies using emotion signals, ^[100] Monitoring public mood for investment decisions ^[101]
Headline ^[61]	News headlines dataset, labeled, regularly updated. ^[102]	News impact analysis on financial markets, ^[103] Automatic headline classification (e.g., positive/negative), ^[104] Event-driven trading system development, ^[105] Detecting market-moving news in real time, ^[106] Training language models for financial headline understanding ^[56]
FiNER-139	Financial named entity recognition open Research Dataset, annotated entities, domain-specific. ^[62]	Financial named entity recognition (NER), ^[107] Extracting companies, ^[108] instruments, and economic terms from text, ^[109] Building financial knowledge graphs, ^[51] Improving document search in finance, ^[110] Automating regulatory compliance monitoring

Machine learning	Description and features	Application
Open-source packages	FNXL ^[111]	Financial news dataset, multilingual, labelled for sentiment. Multilingual financial sentiment analysis, ^[112] Cross-lingual market news understanding, ^[113] Training models for global financial news monitoring, ^[114] Building sentiment-based investment signals, ^[45] Studying sentiment transfer across languages in finance ^[115]
	FinTabNet ^[63]	Financial table dataset, annotated structure, extracted from reports. ^[46] Table structure recognition in financial documents, ^[116] Automatic extraction of financial figures from reports, ^[117] Training models for document digitization, ^[118] Enhancing information retrieval from tabular data ^[119]
	Scikit-learn ^[64]	Machine learning library, easy-to-use, classification, regression, clustering. ^[120] Building classification models, Regression analysis, ^[121] Clustering and unsupervised learning, ^[122] Data preprocessing and transformation, ^[123] Model evaluation and validation, ^[124] Feature selection and dimensionality reduction ^[125]
	TensorFlow ^[67]	Deep learning framework, flexible, scalable, supports neural networks. ^[126] Building deep neural networks, ^[127] Image and speech recognition, ^[128] Natural language processing, ^[129] Time series forecasting, ^[19] Reinforcement learning, ^[130] Large-scale machine learning deployment
	PyTorch ^[65]	Deep learning library, dynamic computation, flexible, popular for research. ^[131] Deep learning research prototyping, ^[132] Computer vision applications, ^[133] Natural language processing tasks, ^[134] Custom neural network development, ^[135] Reinforcement learning experiments, GPU-accelerated model training ^[136]
	Keras ^[66]	High-level neural networks API, user-friendly, runs on TensorFlow. ^[137] Rapid prototyping of neural networks, ^[138] Image classification tasks, ^[139] Text and sentiment analysis, Sequence modelling (RNN, LSTM), ^[140] Transfer learning experiments, Educational deep learning tutorials ^[141]

research data, and the Google Dataset^[59] facilitates the discovery of datasets for market analysis and academic research. For developing sentiment analysis models, datasets such as StockEmotions^[60] provide news articles labeled with emotional indicators, while others like Headline^[61] offer curated news headlines for studying market impact. Furthermore, domain specific resources for named entity recognition, such as FiNER-139,^[62] and for table structure recognition, like FinTabNet,^[63] are essential for automating information extraction from unstructured financial documents.

The development of sophisticated AI models in finance is heavily reliant on powerful and accessible open source software packages. Foundational libraries such as Scikit-learn^[64] offer a user friendly interface for implementing a wide array of machine learning algorithms, including classification, regression, and clustering, making it a primary tool for building predictive models in areas like credit scoring. For more complex deep learning applications, frameworks like TensorFlow provide a flexible and scalable environment for constructing and deploying large neural networks for tasks such as time series forecasting and natural language processing. Similarly, PyTorch^[65] has gained significant popularity, particularly in the research community, for its dynamic computational graph and intuitive interface, which facilitates rapid prototyping in computer vision and

reinforcement learning experiments. High level application programming interfaces like Keras,^[66] which runs on top of frameworks like TensorFlow,^[67] further simplify the process of building and experimenting with neural networks, making deep learning more accessible for a broader range of applications including sentiment analysis and sequence modeling.

2.3 Artificial intelligence models in fintech

Artificial Intelligence now underpins core decisions across lending, payments, trading, insurance, and compliance.^[142] In supervised learning for retail and small business credit, gradient boosted trees, support vector machines, and deep nets routinely outperform traditional scorecards on accuracy while enabling granular risk segmentation at scale.^[143] Benchmarking studies over diverse credit datasets showed robust gains from modern classifiers and careful model selection.^[11] Early evidence from bank portfolios also demonstrated the practical lift of machine learning in default prediction and line management.^[14] New work integrates alternative data and careful sampling to improve performance without eroding governance standards,^[15] while fairness aware development is becoming standard practice, with frameworks to quantify and mitigate disparate impact in credit scoring and to evaluate fairness trade offs alongside profitability.^[144,145]

Explainability for regulated credit decisions is advancing through techniques that surface global and local drivers, which can be embedded into validation and audit processes.^[33]

Payment fraud and financial crime analytics rely on supervised learning, cost sensitive learning, and representation learning on highly imbalanced data.^[146] Classic studies in card fraud detection compared logistic regression, neural networks, and tree ensembles, and motivated practices such as transaction aggregation and cost based evaluation.^[147] Recently, graph neural networks exploit account and device relationships to detect coordinated rings, and are being adopted for transaction monitoring in anti money laundering where scalability and regulatory auditability are essential.^[35] These models often combine supervised detection with unsupervised anomaly scoring to prioritize investigation queues and reduce false positives.

In markets, predictive modeling spans supervised learning on engineered features, deep sequence models for limit order books and returns, and reinforcement learning for policy search in trading and portfolio allocation. Deep networks and boosted trees delivered significant improvements for return prediction and statistical arbitrage in liquid equities.^[27,35] Recurrent and convolutional architectures capture temporal and cross asset structure, with applications to foreign exchange and volatility modelling.^[148,149] For portfolio construction, deep learning is used either to forecast returns that feed optimization or to learn allocation policies directly, with evidence of improved risk adjusted performance in controlled studies and backtests.^[150,151] Reinforcement learning is also advancing in portfolio selection and execution, where reward shaping and risk constraints can be encoded into the objective.^[152]

Natural language processing has moved from dictionaries to transformer based models that read news, filings, and transcripts.^[153] Today, deep models process news streams, earnings call text, and even audio to extract sentiment and forward looking signals that feed trading and credit early warnings.^[154] Large language models are being evaluated for investment analysis and screening, with peer reviewed evidence that model outputs can correlate with subsequent fundamentals and returns, while performance varies with task design and evaluation rigor.^[155] For long documents such as financial reports, neural summarization and information extraction are being applied to accelerate analysis and support compliance reviews.^[156]

Insurance has become a rich testbed for tabular deep learning, telematics analytics, and interpretable pricing. Usage based motor insurance now integrates telematics signals with traditional factors, with studies showing material gains in

claim frequency and severity prediction, and practical pipelines that blend convolutional encoders with generalized linear models to preserve transparency and calibration.^[157,158] Actuarial journals also document advances in model calibration and risk adjusted evaluation for pricing, reserving, and reinsurance decisions, including deep learning solutions to optimal control problems in capital management.^[159,16]

Across these domains, explainability, robustness, and fairness are central to deployment. Financial institutions must reconcile model performance with attributable and responsible analytics, and the literature offers frameworks for explainable artificial intelligence in operations research and finance, alongside perspectives that advocate inherently interpretable models for high stakes decisions.^[161,162]

Table 2 summarized the principal artificial intelligence algorithms applied in fintech and their advantages. Large language models facilitated natural language tasks such as financial news summarization, sentiment analysis, and chatbots. Neural networks and LSTMs supported credit scoring, stock prediction, and volatility modeling, while CNNs addressed image-based verification tasks. Random forests and decision trees provided interpretable models for credit and risk assessment. Explainable AI enhanced transparency, fusion models combined algorithmic strengths, generative AI enabled synthetic data and reporting, and reinforcement learning optimized trading and portfolio strategies.

Table 2 clarified that artificial intelligence models in fintech were not interchangeable, because each model class offered different technical strengths and matched different financial tasks. The comparison showed that model suitability depended mainly on the form of the input data, the degree of temporal dependence, the required balance between predictive accuracy and interpretability, and the practical objective of the institution. First, the table showed a clear distinction between text oriented models and structured numerical models. Large language models were particularly suitable for tasks involving unstructured textual information, such as financial news summarization, chatbot services, sentiment analysis, report generation, and fraud email detection. Their main advantage lay in natural language understanding and generation, which made them more appropriate for communication, document processing, and language based intelligence tasks than for conventional tabular risk prediction. By contrast, neural networks and random forest models were more naturally aligned with structured financial datasets used in credit scoring, default prediction, fraud detection, and market trend analysis.

Neural networks, LSTM models, and reinforcement

Table 2: Artificial intelligence models and its core advantages for fintech applications.

Artificial Intelligence Algorithm	Key strengths	Fintech applications
Large language model ^[163]	Excels at understanding and generating natural language text ^[164]	Financial news summarization, ^[165] chatbots for customer service, ^[166] sentiment analysis, ^[167] report generation, ^[168] fraud email detection ^[165]
Neural Networks ^[169]	Powerful nonlinear modeling, adapts well to complex data ^[170]	Credit scoring, ^[171] stock price prediction, ^[172] loan default risk, ^[173] pattern recognition in trades, ^[174] asset price forecasting ^[175]
Long short term memory (LSTM) ^[176]	Captures long-term dependencies, strong for sequential data processing ^[177]	Time-series forecasting, ^[178] stock price prediction, ^[179] market volatility modelling, ^[180] sequence anomaly detection, ^[181] economic indicator prediction ^[182]
Convolutional neural network ^[183]	Excellent at image recognition, efficient feature extraction ^[184]	Document image analysis, ^[185] cheque signature verification, ^[186] chart pattern recognition, ^[187] fraudulent document detection, ^[42] visual compliance screening ^[188]
Random forest ^[189]	Strong against noise, handles nonlinear classification effectively ^[190]	Credit approval, ^[191] fraud detection, ^[192] customer segmentation, ^[193] loan risk assessment, ^[194] market trend analysis ^[195]
Decision tree ^[196]	Simple, interpretable results with fast computation speed ^[197]	Credit decisioning, ^[198] loan approval, ^[199] customer churn prediction, ^[200] simple risk classification, ^[196] transaction categorization ^[201]
Explainable AI (XAI) ^[202]	Improves transparency, enhances trust in AI systems ^[203]	Transparent credit scoring, ^[36] regulatory compliance, ^[204] model auditability, ^[205] bias detection, ^[206] explain loan decisions ^[207]
Fusion models	Combines model strengths, improves prediction accuracy significantly ^[208]	Multi-source risk analysis, ^[209] ensemble trading strategies, ^[210] cross-market forecasting, ^[211] hybrid fraud detection, ^[212] integrated portfolio analytics ^[213]
Generative AI ^[214]	Produces creative outputs, enables innovative application scenarios ^[215]	Synthetic data generation, ^[216] scenario simulation, ^[217] automated report drafting, ^[218] financial content creation, ^[219] market scenario modelling ^[220]
Reinforcement Learning (RL) ^[221]	Learns from feedback, optimizes long-term decision strategies ^[222]	Algorithmic trading, ^[18] portfolio optimization, ^[223] asset allocation, ^[41] dynamic hedging, ^[224] market making ^[225]

learning were better suited to tasks characterized by complexity, dynamic interaction, or time dependence. General neural networks were effective when financial relationships were highly nonlinear, such as in credit scoring, asset pricing, or trade pattern recognition. LSTM models were more specifically suited to sequential and time series problems because they captured long term temporal dependencies. This made them more appropriate for stock prediction, volatility modelling, economic indicator prediction, and anomaly detection in sequential market data. Reinforcement learning differed from both by focusing on sequential decision optimization under feedback. For that reason, it was more suitable for algorithmic trading, portfolio optimization, asset allocation, dynamic hedging, and market making, where actions affected future outcomes and long horizon strategy mattered.

Decision trees and explainable AI methods were especially valuable when institutions needed transparent reasoning, auditability, and regulatory acceptance. Decision trees were useful in credit decisioning, loan approval, and transaction categorization because they produced simple and understandable decision rules with low computational burden. Explainable AI played a different but complementary role. It did not merely improve prediction, but strengthened trust,

model governance, bias detection, and compliance. This made it particularly relevant in high stakes applications such as credit scoring and regulated lending, where institutions needed to justify automated decisions to regulators and customers.

Convolutional neural networks were more appropriate for visual financial tasks such as document image analysis, cheque signature verification, chart pattern recognition, and fraudulent document detection. Their strength in feature extraction from image based inputs distinguished them from tree based methods or language models, which were not designed primarily for visual pattern recognition.

Fusion models combined the strengths of multiple algorithms and were therefore suitable for multi source risk analysis, hybrid fraud detection, and integrated portfolio analytics, where no single model could fully capture the complexity of the task. Generative AI was more suitable for synthetic data generation, scenario simulation, automated drafting, and financial content creation because it produced new outputs rather than only making classifications or forecasts. This suggested that its role in fintech extended beyond prediction into simulation, augmentation, and workflow support.

Overall, the choice of artificial intelligence model in fintech was task dependent rather than universal. Models

differed in their strengths because they were designed for different forms of data and decision problems. Large language models were more suitable for language intensive tasks, convolutional neural networks were more suitable for visual analysis, LSTM models were more suitable for temporal forecasting, reinforcement learning was more suitable for dynamic optimization, and decision trees and explainable AI were more suitable when interpretability and compliance were essential. Neural networks and random forest models remained broadly useful for nonlinear prediction in structured financial settings, while fusion models offered advantages when complex applications required multiple complementary capabilities. Therefore, certain model categories were better suited to specific financial tasks because their technical properties aligned more closely with the informational, operational, and regulatory demands of those tasks.

2.4 AI applications in fintech

Artificial intelligence has become a transformative force across the financial technologies industry, enabling a wide spectrum of applications that enhance operational efficiency, improve customer experiences, and strengthen risk management frameworks.^[226] In Fig. 2, AI models are being deployed to automate and optimize core financial tasks, from client interaction and security to complex data analysis and decision making processes. The integration of AI is not merely an incremental improvement but a fundamental shift in how financial services are designed, delivered, and managed, touching virtually every aspect of the fintech ecosystem.

AI has profoundly reshaped customer facing financial services. Table 3 summarizes the diverse applications of artificial intelligence across fintech services, detailing how technologies from voice recognition to generative AI agents deliver significant benefits such as enhanced security, operational efficiency, and personalized customer experiences.

Biometric technologies, including voice and facial recognition, are increasingly used for secure identity verification and authentication, streamlining customer onboarding and transaction authorization while reducing reliance on traditional passwords.^[227] In the domain of customer support, Natural Language Processing (NLP) underpins the functionality of sophisticated chatbots and virtual assistants.^[228] These AI driven tools provide 24 for 7 support, handle routine inquiries, and guide users through complex processes, thereby improving service quality and reducing operational costs for financial institutions. Furthermore, AI algorithms are central to the delivery of personalized financial services. Robo advisors leverage machine learning to construct and manage investment portfolios tailored to individual risk profiles and financial goals, making sophisticated wealth management accessible to a broader audience.^[37] These systems also power personalized recommendation engines that suggest relevant financial products, from credit cards to insurance plans, based on a user's behavior and financial data.

In credit underwriting and risk scoring field, supervised learning models have outperformed traditional scorecards in benchmark studies, particularly on imbalanced default data and for small and mid size enterprise lending.^[229] These systems exploit nonlinear interactions among bureau, transactional, and alternative features to improve rank ordering and cut loss given default through earlier interventions. Recent work focuses on fairness and interpretability, showing that monotone gradient boosted trees and rule based surrogates can preserve accuracy while providing case level explanations suitable for audit and consumer disclosure.^[145,230,231] Empirical evidence also shows performance gains when combining feature learning with human readable constraints.

Machine learning models are extensively used for the real

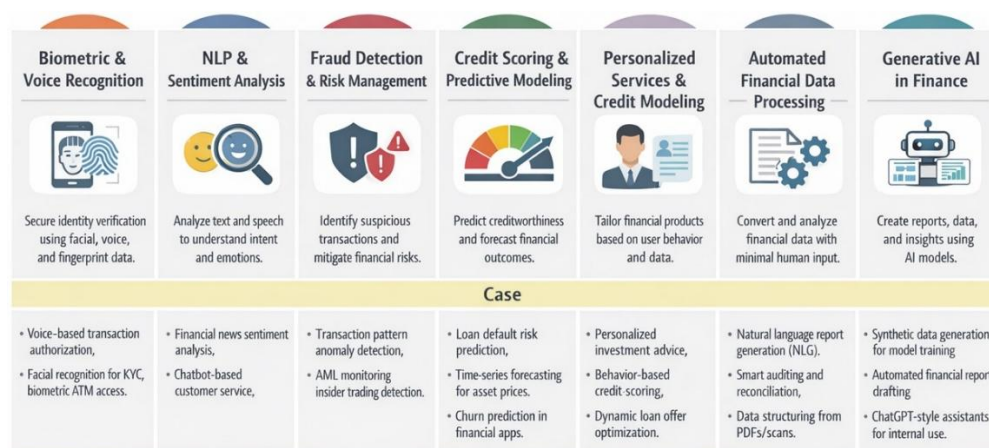


Fig. 2: AI applications in fintech industry.

Table 3: Applications and benefits of artificial intelligence in the fintech service.

AI applied field	Applications	Benefit and advantages of AI in finance
Voice recognition ^[235]	Biometric Authentication, ^[236] Voice Banking, ^[237] Transaction Authorization ^[238]	Providing secure, hands-free authentication, streamlining transactions, and improving customer verification with greater speed and efficiency
Sentiment analysis ^[239]	Investment prediction, ^[240] chat sentiment analysis, ^[241] Brand, Voice-based Emotion Detection ^[242]	Sentiment analysis helps financial services anticipate trends, enhance customer experience, and manage risks through emotional insight and real-time feedback.
Cheating, criminal detection	Fraud Detection in Transactions, ^[243] Money Laundering Monitoring (AML), ^[53] Insider Trading Surveillance, ^[244] Document Forgery Detection ^[243]	AI enables real-time detection of fraud, deception, and financial crimes, enhancing security and regulatory compliance in financial services. ^[245]
Customized recommendation	Personalized Investment Advice, ^[246] Tailored Product Offers, ^[247] Spending Insights and Alerts, Financial Planning Tools, ^[248] Behavior-Based Rewards Programs ^[249]	Delivers relevant, user-specific financial products and insights efficiently.
Financial information process	Automated Financial Statement Generation, ^[44] Smart Auditing and Reconciliation, ^[250] Data Structuring from Unstructured Sources, ^[251] Natural Language Report Generation (NLG), ^[252] Credit Scoring Enhancement ^[15]	Automating reporting, enhancing accuracy, enabling real-time insights, and ensuring regulatory compliance efficiently.
Image identification	KYC Document Verification, ^[227] Check Image Processing, ^[253] ATM Fraud Detection, ^[254] Signature Verification, ^[255] Receipt and Invoice Scanning ^[256]	Speeds verification, enhances security, and automates visual data processing.
Customer service	AI-Powered Voice Support (IVR), ^[257] Chatbots, ^[22] Automated Ticket Routing, ^[258] Multilingual Customer Support, ^[259] Customer Satisfaction Analysis ^[260]	Personalized support with reduced human workload.
Predictive modelling	Credit Risk Assessment, ^[10] Customer Churn Prediction, ^[261] Operational Risk Management, ^[262] Loan Pre-approval Automation, ^[263] Personalized Product Recommendations ^[264]	Enables smarter decisions, risk reduction, and business growth forecasting.
Information security	Intrusion Detection Systems (IDS), ^[265] Phishing Email Detection, ^[266] Anomaly Detection, ^[267] User Authentication Enhancement, ^[268] Malware Detection and Prevention ^[269]	Enhances protection, rapid response, and adaptive threat detection.
Generative artificial intelligence agent	Automated Report Generation, ^[270] Synthetic Data Generation, ^[271] Personalized Marketing Content, ^[272] Conversational AI (Chatbots/Assistants), ^[273] Code Generation and Automation ^[274]	Boosts efficiency, creativity, and personalization in digital financial services.

time detection and prevention of fraudulent activities. By analyzing vast streams of transaction data, these systems can identify anomalous patterns indicative of credit card fraud, identity theft, or account takeovers with high accuracy.^[16,42] Beyond transactional fraud, AI is a crucial tool in the fight against financial crime. Advanced algorithms are deployed for Anti Money Laundering (AML) compliance to monitor transactions, detect suspicious networks of activity, and reduce the high volume of false positives often associated with legacy rule based systems.^[232] The predictive capabilities of AI also extend to comprehensive risk management, where models are used to forecast market volatility, assess operational risks,

and conduct stress tests on financial portfolios to ensure institutional resilience.^[33]

AI is also instrumental in automating the complex and labor intensive processes of financial data analysis and reporting. NLP and computer vision technologies are used to automatically extract and structure information from unstructured documents like financial statements, contracts, and news articles.^[233] Sentiment analysis models process news, social media, and earnings call transcripts to gauge market mood and provide predictive signals for investment strategies.^[234] The emergence of generative AI has introduced further advancements. Large language models are now

capable of generating sophisticated outputs, including automated financial report summaries, personalized marketing content, and synthetic data for training more robust machine learning models without compromising customer privacy.^[52] These generative tools are poised to further automate knowledge work and unlock new efficiencies across the financial industry.

3. AI agents in fintech

The evolution of artificial intelligence in the financial sector is marked by a significant transition from task specific models to autonomous AI agents.^[275] AI agents were particularly suitable for financial tasks that required autonomous decision support, multi step workflow execution, and real time interaction with dynamic information. In fintech settings, they were especially appropriate for intelligent customer service, personalized financial advisory, automated compliance monitoring, fraud investigation, portfolio rebalancing support, and trading assistance, because these tasks involved continuous information gathering, context aware reasoning, and adaptive responses rather than single step prediction alone. Compared with conventional models that focused mainly on classification or forecasting, AI agents were more useful when the system needed to combine language understanding, tool use, rule based checking, and sequential decision making within one process. For this reason, AI agents were better suited to complex financial tasks that required coordination across data sources, repeated feedback, and operational action, particularly in areas such as customer interaction, risk monitoring, regulatory reporting, and decision support for investment management. These agents are capable of sensing their environment, making independent decisions, executing actions to achieve predefined goals with minimal human intervention and learn from feedback, as shown in Fig. 3. This progression represents a move from tools that provide analysis to autonomous entities that perform complex knowledge work, fundamentally reshaping financial operations. An AI agent's architecture typically uses a large language model as a reasoning engine, augmented with specialized tools, memory, and planning capabilities, allowing it to perform multi step tasks that were previously infeasible for a single model.^[276]

AI agents are being deployed across a spectrum of financial applications, driving innovation and efficiency. In personalized finance, they function as autonomous financial assistants, offering services that range from customized investment advice to dynamic portfolio management. These agents analyze user data and market trends to provide tailored recommendations, making sophisticated financial guidance more accessible. In the domain of algorithmic trading, AI

agents introduce a new level of sophistication by autonomously gathering market intelligence, developing strategies, and executing trades.^[278] Reinforcement learning techniques are often employed to train these agents, enabling them to learn and refine their strategies through continuous interaction with market environments.^[279] This allows them to adapt to market volatility and optimize for risk adjusted returns in ways that surpass traditional algorithmic methods.

Furthermore, the concept of multi agent systems is gaining traction, where multiple specialized agents collaborate to solve complex financial problems.^[280] These systems can simulate real world market dynamics by assigning different roles to various agents, such as fundamental analysts, sentiment analysts, and risk managers.^[281] Through structured debate and collaboration, these agents can reduce cognitive biases and improve the robustness of investment decisions. Multi agent frameworks are being explored to enhance market analysis, manage portfolios, and even stress test financial strategies within simulated environments, providing deeper insights than single agent systems can offer.^[6] In Fig. 4, a multi agent classifies the user query and routes it to a credit agent or a fraud agent, which call scoring and anomaly models with transaction and profile data to return an approval or fraud suspicion as the final answer.

The application of AI agents also extends to critical back office and compliance functions. In anti money laundering and fraud detection, agents can automate the investigation of suspicious activities by gathering data from various systems, analyzing patterns, and even generating suspicious activity reports for human review.^[282] This automation of compliance workflows helps financial institutions manage operational risks, reduce manual workloads, and ensure adherence to complex regulations. However, the deployment of these autonomous systems introduces significant challenges. Ensuring reliability, preventing erroneous or malicious actions, and maintaining transparency are critical concerns that require robust governance frameworks.^[283] The "black box" nature of some advanced models necessitates the development of explainable AI (XAI) to ensure that the decisions made by agents are interpretable and auditable by regulators and stakeholders.^[8] The successful integration of AI agents in fintech will depend on balancing their innovative potential with stringent security protocols and a commitment to responsible, human supervised deployment.

4. Challenges of AI in fintech

The integration of artificial intelligence into the financial technology sector has catalyzed a paradigm shift, offering unprecedented efficiencies in areas like algorithmic trading,

What Are AI Agents?

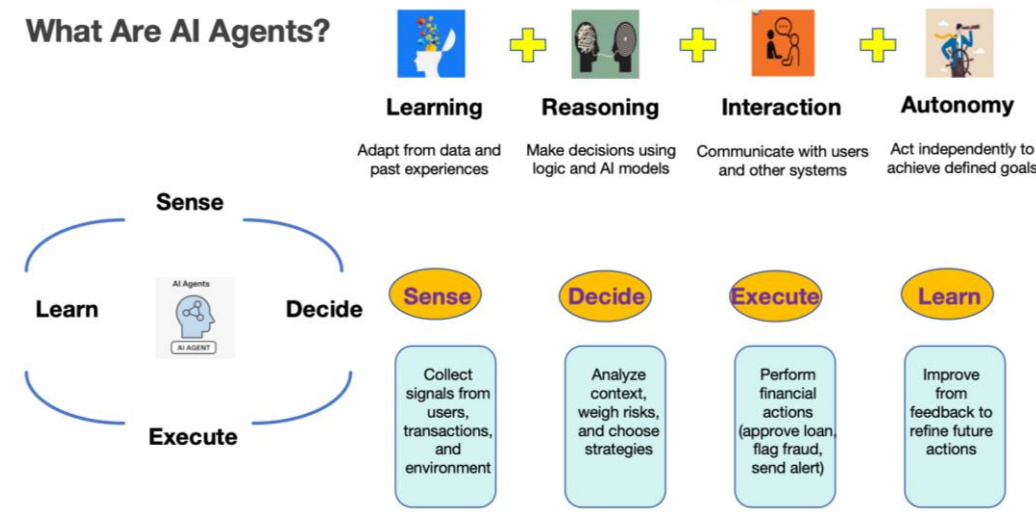


Fig. 3: Core architecture and functional components of AI agents in fintech.

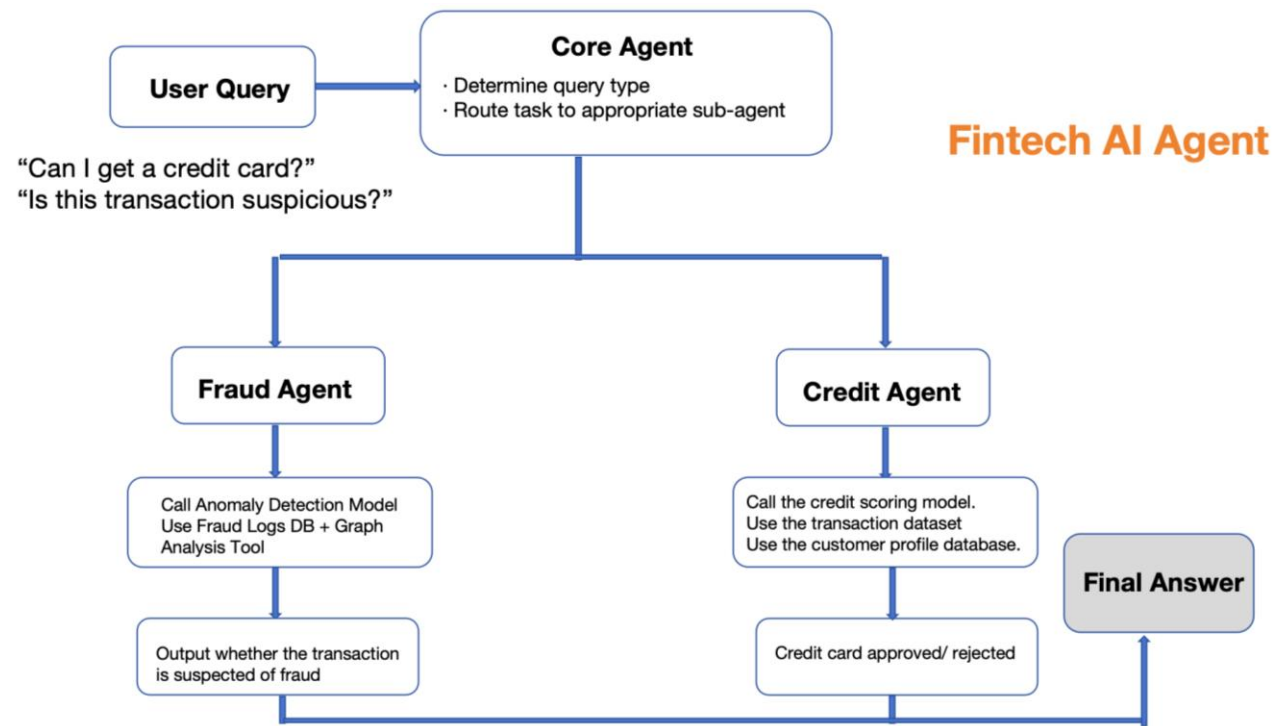


Fig. 4: Multi-agent workflow for credit assessment and fraud detection in fintech.

credit scoring, and fraud detection.^[284] However, as shown in Fig. 5, the deployment of these sophisticated systems introduces a unique set of technical challenges that require rigorous scientific inquiry to overcome. These challenges span the entire lifecycle of an AI model, from data acquisition and model development to deployment and ongoing governance.^[285] Addressing these issues is paramount for ensuring the creation of robust, fair, and trustworthy AI powered financial systems.^[286] This section reviews the primary technical obstacles confronting the application of AI

in the fintech industry, focusing on data related issues, model interpretability, security vulnerabilities, algorithmic fairness, and regulatory compliance.

A foundational challenge for AI in fintech revolves around data quality, availability, and privacy.^[287] Financial datasets are often characterized by significant noise, missing values, and non stationary distributions, which can severely degrade the performance of machine learning models.^[288] For instance, stock market data is notoriously volatile and subject to sudden regime changes, making historical data a potentially

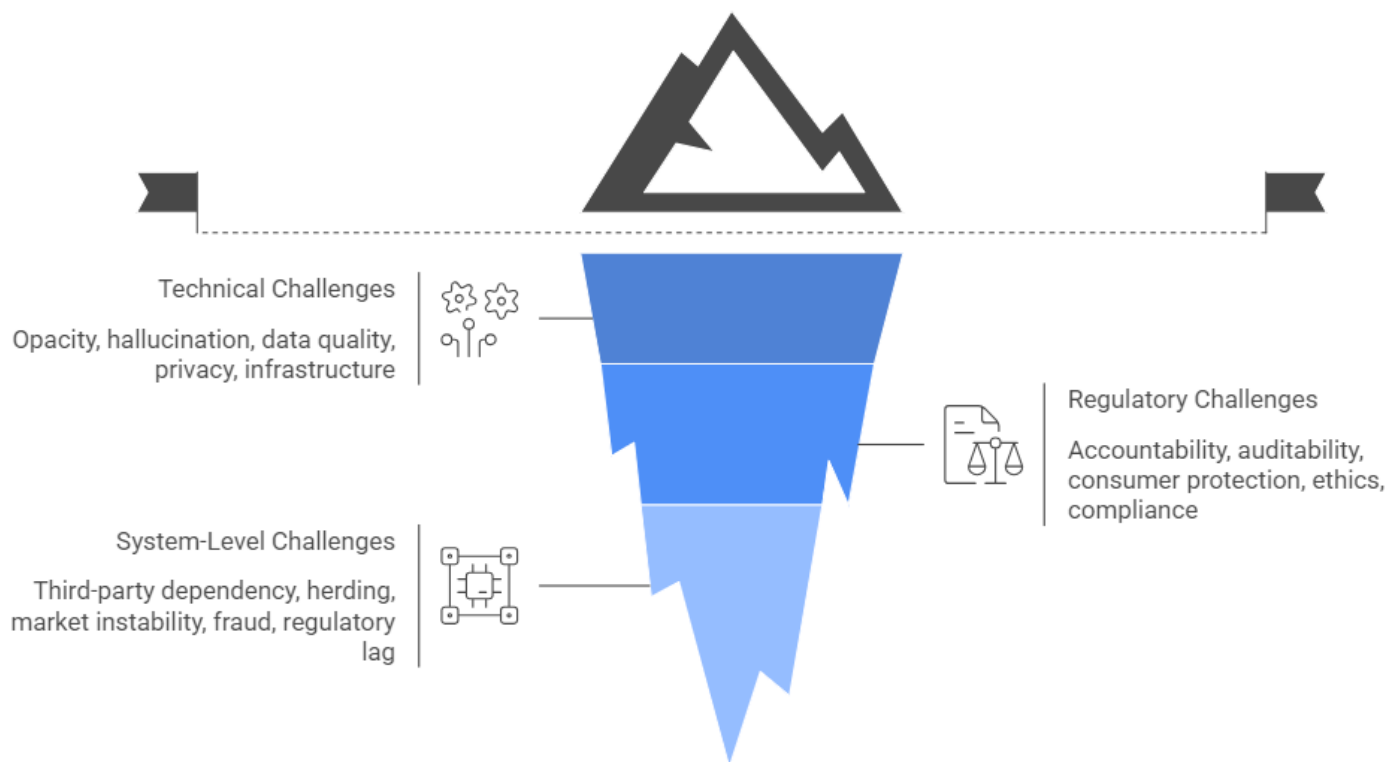


Fig. 5: Technical, regulatory and system challenges of AI deployment in fintech.

unreliable predictor of future performance. Furthermore, the most valuable financial data, such as individual transaction histories, is protected by stringent privacy regulations like the General Data Protection Regulation.^[289] This creates a complicated environment for data acquisition and sharing. Techniques such as federated learning^[290] have been proposed to train models on decentralized data without compromising user privacy, yet these methods introduce their own complexities related to communication overhead and statistical heterogeneity. The successful application of AI is therefore heavily dependent on advanced data preprocessing techniques and privacy preserving machine learning architectures.^[291]

The most widely discussed technical challenge is the lack of transparency in complex AI models, often termed the ‘black box’ problem.^[292] Many high performing models, such as deep neural networks, operate in a way that is not readily understandable to human. In the high stakes environment of finance, this opacity is unacceptable to regulators, customers, and internal risk managers who need to understand the rationale behind automated decisions, such as a loan application denial decision. This necessity has spurred the growth of eXplainable AI, or XAI, a field dedicated to developing techniques that can render model decisions interpretable.^[202] Methods like Shapley Additive Explanations (SHAP)^[293] and Local Interpretable Model agnostic Explanations (LIME)^[294] are gaining traction, their application

in complex financial scenarios is still an active area of research. The challenge is to provide meaningful explanations without sacrificing the predictive accuracy that makes these models valuable in the first place.

Model overfitting is another challenge. This occurs when a machine learning model learns the training data too precisely, capturing not only the underlying patterns but also the noise and random fluctuations specific to that dataset.^[295] While such a model may exhibit excellent performance during testing on historical data, its ability to generalize to new, unseen data is severely compromised. In finance, the consequences can be catastrophic. An overfitted algorithmic trading strategy may perform exceptionally in backtests only to incur substantial losses when deployed in live markets.^[296] The technical challenge involves implementing robust validation techniques beyond simple data splitting. Methods such as k fold cross validation, walk forward optimization, and strict regularization are essential for building models that demonstrate genuine predictive power rather than a mere ability to memorize the past.^[297]

With the rise of large language models and other generative AI, the problem of hallucination has emerged as a critical concern. Hallucination refers to the tendency of these models to generate outputs that are nonsensical, factually incorrect, or entirely fabricated, yet are presented with a high degree of confidence. In a fintech context, a customer service chatbot might invent a nonexistent financial product, or an

Table 4: Challenges and limitations of artificial intelligent applied in fintech.

Applied Field	Challenges	Applications	Limitations
Voice recognition ^[304]	Accents and noisy input	Voice-activated banking, ^[305] phone payment authentication, ^[306] customer identity verification, ^[307] voice-controlled chatbots, ^[308] fraud call detection ^[309]	Accent sensitivity, ^[310] background noise interference, ^[311] privacy concerns, ^[312] limited language support, ^[313] error in voice matching ^[314]
Sentiment analysis ^[315]	Misinterpretation of emotions	Social media monitoring, ^[1] news-based trading signals, ^[316] customer feedback analysis, ^[234] product review mining, ^[317,318] market mood tracking	Sarcasm detection difficulty, ^[319] context misunderstanding, ^[320] language limitations, ^[321] data labelling cost, ^[322] overfitting to text style ^[322]
Cheating, criminal detection ^[232]	Evolving fraud and crime tactics	Suspicious transaction alerts, ^[320] identity theft detection, ^[323] insider trading surveillance, ^[324] money laundering screening, ^[325] synthetic identity flagging ^[326]	High false positives, ^[320] privacy intrusion, ^[327] evolving tactics, ^[328] data shortage, ^[329] legal restrictions ^[330]
Customized recommendation ^[331]	Incomplete user profiles	Personalized product offers, ^[331] tailored investment portfolios, ^[332] credit card spend suggestions, ^[333] loan product targeting, ^[334] insurance plan customization ^[331]	Filter bubble risk, ^[335] user privacy issues, ^[327] cold start problem, ^[336] over-personalization, ^[337] limited cross-channel data ^[338]
Fraud detection ^[320]	Changing fraud patterns	Real-time transaction screening, ^[339] credit card fraud alerts, account takeover detection, ^[340] insurance claim verification, ^[341] payment gateway monitoring ^[342]	Adaptive fraud tactics, ^[343] model drift, ^[1] imbalanced data, ^[344] latency in detection, ^[345] high computational cost ^[346]
Credit scoring ^[347]	Biased or incomplete data	Loan eligibility assessment, ^[348] dynamic credit limit setting, ^[349] small business loan evaluation, ^[350] peer-to-peer lending risk ^[351]	Biased training data, ^[352] lack of transparency, ^[347] limited data sources, ^[13] regulatory barriers, ^[353] explainability issues ^[347]
Algorithmic trading ^[18]	High market volatility	Automated buy/sell execution, ^[354] arbitrage strategy deployment, ^[355] trend-following bots, ^[356] high-frequency trading, ^[357] portfolio rebalancing ^[358]	Overfitting models, ^[320] flash crash risk, ^[357] high transaction costs, ^[359] latency sensitivity, market regime changes ^[360]
Robo-advisory ^[37]	Low user trust	Automated portfolio management, ^[361] risk profiling for clients, ^[362] tax optimization strategies, ^[363] goal-based investment planning, ^[364] retirement savings automation ^[365]	Limited personalization, ^[366] regulatory constraints, ^[367] user trust issues, ^[368] lack of human insight, ^[369] data dependency ^[1]
Risk management ^[370]	Complex risk factors	Market risk forecasting, ^[335] credit analysis, ^[371] stress testing portfolios, ^[372] liquidity risk monitoring, ^[373] operational risk detection ^[374]	Incomplete data, ^[320] scenario coverage limits, ^[375] stress test assumptions, ^[376] response lag ^[377]
Customer service automation ^[378]	Limited language understanding	AI chatbots for inquiries, ^[379] 24/7 virtual assistants, ^[228] automated complaint resolution, ^[380] loan application support, ^[381] transaction status updates ^[382]	Limited empathy, ^[383] misunderstanding intent, ^[384] language limitations, ^[385] escalation issues, ^[386] data privacy concerns ^[303]

automated market analysis tool could generate a report based on hallucinated economic data, leading to misguided investment decisions.^[298] The core technical challenge is to ensure the factual grounding of generative models. Techniques such as retrieval augmented generation (RAG),^[299] which enable a model to retrieve and incorporate information from a verified knowledge base before generating a response, represent a promising direction for mitigating this significant risk.

Security is another critical technical concern. As financial

institutions increasingly rely on AI for core functions, these systems become attractive targets for adversarial attacks.^[300] An adversarial attack involves making small, often imperceptible, perturbations to a model's input data with the goal of inducing an incorrect output. In fintech, this could manifest as manipulating input data to have a fraudulent transaction approved or to trigger a desirable outcome in an algorithmic trading system.^[301] AI models in finance and other domains faced adversarial attacks, data poisoning, model inversion, and evasion tactics that reduced reliability and

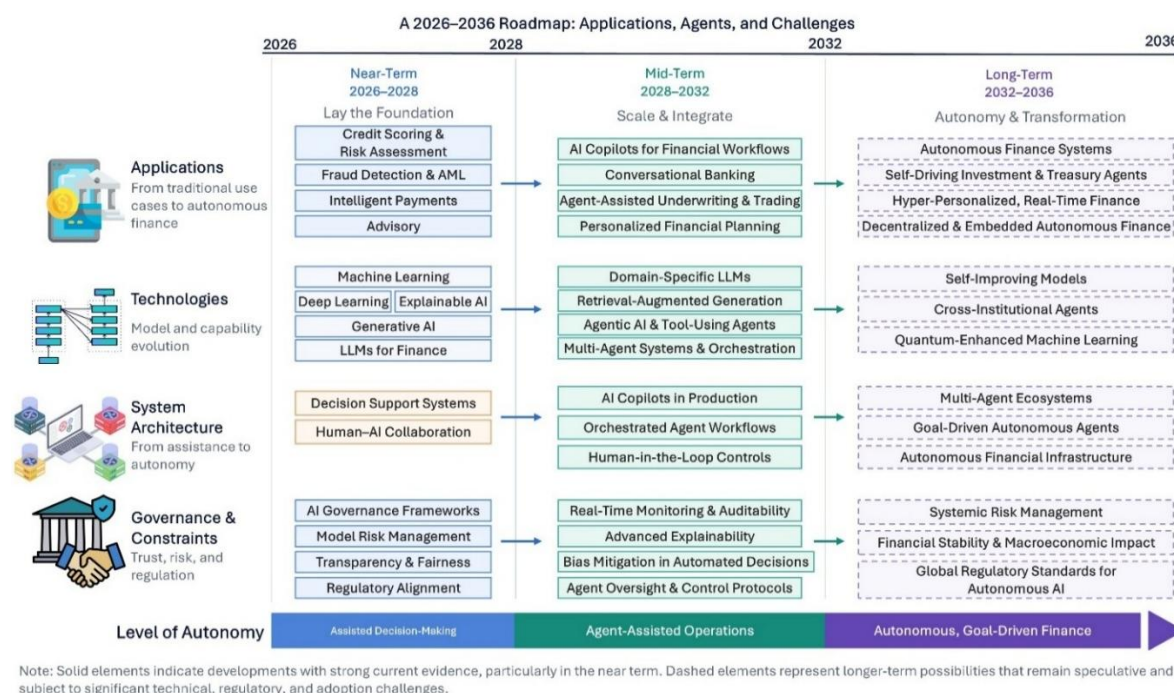


Fig. 6: A conceptual roadmap from 2026–2036 of AI in fintech across applications, technologies, system architecture, and governance constraints. Solid elements represent developments supported by current empirical or industry evidence, while dashed elements indicate forward looking or more speculative trajectories. Arrows denote directional evolution and increasing system autonomy across time.

exposed sensitive information. The main challenges lay in the high dimensionality of input data, the opacity of deep models, the dynamic nature of adversaries, and the trade off between robustness and accuracy. Defences included adversarial training to expose models to perturbed inputs, differential privacy and regularization to limit information leakage, anomaly detection to flag manipulated queries and monitoring with human in the loop oversight. Despite progress, defending AI models required continuous adaptation since attackers evolved strategies quickly and traditional static defenses often degraded under novel threats.

Furthermore, the potential for AI models to perpetuate and even amplify existing societal biases is a significant ethical and technical challenge.^[302] AI systems learn from historical data, and if this data reflects past discriminatory practices, the resulting models will inherit these biases. In fintech applications like credit scoring, this can lead to unfair or discriminatory outcomes for certain demographic groups, even if sensitive attributes like race or gender are excluded from the model's input. Ensuring fairness requires careful auditing of data and models, as well as the development of fairness aware machine learning algorithms.^[145] More specifically, AI systems trained on historical or socially derived data could inadvertently reproduce structural inequities when the underlying data reflected patterns of exclusion or discrimination. In applications such as credit

scoring and fraud detection, this made bias auditing, the use of representative data, and fairness aware model design essential for reducing the risk of systematically unfair outcomes across demographic and socioeconomic groups. Importantly, fairness was not defined by a single criterion. Demographic parity focused on whether positive outcomes were distributed at similar rates across groups, whereas equalized odds focused on whether error rates were balanced conditional on actual outcomes. In financial practice, these definitions often created trade offs because a model that satisfied one fairness objective did not necessarily satisfy another, particularly when default risk distributions differed across groups. Empirical studies in credit scoring and credit ratings showed that fairness interventions could reduce disparate impact, but they could also affect predictive accuracy, profitability, and approval policies, which highlighted a practical fairness versus performance trade off in regulated financial settings.^[145] For this reason, implementing a system that was both accurate and demonstrably fair according to legal and ethical standards remained a major technical hurdle for the industry.

The dynamic and stringent regulatory landscape presents an ongoing technical challenge for AI adoption in fintech. Financial institutions must ensure their AI systems comply with a complex web of regulations governing data privacy, model risk management, and consumer protection.^[303] Translating these legal and ethical principles into concrete

technical specifications for AI systems is not straightforward. This requires the development of new frameworks for AI governance, model validation, and continuous monitoring to ensure that systems operate within prescribed boundaries over their entire lifecycle.^[142]

AI driven financial decision making also raised substantive ethical and governance concerns beyond technical performance alone. The review showed that algorithmic bias remained a major risk because models learned from historical financial data that could encode past discrimination, which in turn could produce unfair outcomes in credit scoring, risk assessment, and customer targeting even when protected attributes were excluded from model inputs. It also showed that accountability and traceability were essential in high stakes financial contexts because institutions had to explain how automated decisions were produced, identify the data and model factors that shaped those outcomes, and maintain auditable records for internal governance and regulatory review. In addition, data governance remained central to responsible deployment. Financial institutions had to address privacy, informed consent, lawful data use, and secure data sharing, particularly when alternative data and cross platform behavioral information were incorporated into model development. These concerns extended to model risk management, where validation, monitoring, and lifecycle controls were needed to detect drift, bias, hallucination, and adversarial vulnerabilities after deployment. Taken together, these issues showed that responsible AI adoption in fintech depended not only on predictive accuracy, but also on fairness, transparency, privacy protection, and continuous compliance with evolving regulatory frameworks.

Table 4 summarized how different AI applications in fintech such as voice recognition, sentiment analysis, fraud detection, credit scoring, algorithmic trading, robo advisory, risk management, and customer service automation faced domain specific challenges and practical limitations despite offering diverse applications across financial services.

5. Conclusion and outlook

Artificial intelligence has moved from being a set of isolated analytical tools to becoming a foundational infrastructure for financial technology. Across lending, payments, fraud detection, trading, wealth management, customer service and compliance, AI systems are reshaping how financial information is processed, how risks are assessed and how financial decisions are delivered. This Review has shown that the field is no longer defined only by conventional machine-learning models for prediction and classification, but increasingly by the convergence of deep learning, natural

language processing, generative AI, autonomous agents and multi-agent systems. These developments are expanding the scope of fintech from task-specific automation towards adaptive, interactive and increasingly autonomous financial workflows. In this context, AI agents are especially important because they connect data interpretation, reasoning, tool use and sequential decision-making, thereby offering a new model for intelligent financial services rather than merely improving existing predictive pipelines.

The long-term value of AI in fintech will depend less on technical performance alone than on whether these systems can be made robust, interpretable, fair, secure and governable in high-stakes financial environments. Persistent challenges remain in data quality and privacy, model opacity, algorithmic bias, adversarial vulnerability, hallucination in generative models, regulatory compliance and human oversight. Future research should therefore move beyond accuracy-centred evaluation and develop integrated frameworks for trustworthy financial AI, including privacy-preserving learning, domain-specific explainability, fairness-aware validation, agent safety, continuous monitoring and regulatory auditability. The next stage of AI-driven fintech will require closer coordination between technical innovation, institutional governance and responsible regulation. If these challenges are addressed, AI has the potential not only to improve efficiency and personalization in financial services, but also to support a more transparent, accountable and resilient financial ecosystem.

A significant shift is underway from general purpose AI models to highly specialized financial foundation models and autonomous agents capable of complex reasoning and decision making.^[387] These autonomous systems are poised to revolutionize core financial processes by operating with minimal human input to perform tasks like dynamic portfolio rebalancing, real time risk analysis, and automated compliance checks.^[388] The development of AI agents that can perceive their environment, analyze vast streams of information, and execute actions to achieve specific financial goals will move the industry closer to a state of autonomous finance. This will enable financial institutions to enhance workflow optimization, improve customer experiences, and strengthen decision making, thereby gaining a significant competitive advantage.

Fig. 6 presents a structured roadmap of the evolution of artificial intelligence in financial technology over the period 2026 to 2036, integrating developments across applications, underlying technologies, system architecture, and governance constraints within a unified analytical framework. The roadmap illustrates a staged transition from predictive and task

specific machine learning systems toward generative AI and, subsequently, agentic and multi-agent financial systems with increasing levels of autonomy. In the near term, developments are primarily grounded in established machine learning, explainable AI, and early generative systems, which augment decision making in areas such as credit assessment, fraud detection, and financial advisory services. The mid term phase is characterized by the scaling of domain specific large language models, retrieval augmented systems, and AI copilots, alongside the emergence of orchestrated agent workflows that extend automation across financial processes. In the long term, a potential shift toward autonomous financial ecosystems enabled by multi-agent coordination and self improving systems, while explicitly recognizing that these trajectories remain constrained by technical, regulatory, and systemic considerations.

The future of customer interaction in fintech will be defined by AI driven hyper personalization, moving far beyond current segmentation and basic customization.^[389] By leveraging real time data analytics and machine learning, financial institutions can create uniquely tailored experiences, products, and advice that align with an individual's specific financial situation, behaviors, and life goals.^[390] This approach allows banks to anticipate customer needs, provide proactive financial guidance, and foster deeper, more meaningful relationships.^[391] AI powered virtual assistants and chatbots will evolve to handle more complex and nuanced interactions, providing personalized support that improves accuracy, accelerates response times, and ultimately enhances customer satisfaction and loyalty.^[392]

As AI systems become more powerful, the imperative to address their inherent challenges with next generation solutions grows stronger. The "black box" problem is being actively countered by progress in explainable AI (XAI), which aims to make model decisions transparent and auditable, a critical requirement for regulatory compliance and building trust.^[393] Research is increasingly focused on developing hybrid and inherently interpretable models that do not sacrifice predictive accuracy for transparency.^[394] Simultaneously, the critical need for data privacy is being met by the maturation of Privacy Enhancing Technologies (PETs).^[395] Techniques like homomorphic encryption,^[396] which allows computation on encrypted data, and federative learning^[397] are enabling collaborative data analysis and model training across institutions without exposing sensitive raw data, fostering innovation while upholding stringent privacy standards.

Looking further ahead, the convergence of AI with other frontier technologies like quantum computing promises to

unlock capabilities previously thought impossible. Quantum Machine Learning (QML)^[398] holds the potential to solve complex optimization problems in portfolio management and risk assessment at speeds and scales unattainable for classical computers. This could revolutionize algorithmic trading and the development of sophisticated financial models. In parallel, generative AI is becoming a powerful tool for creating highly realistic financial market simulations.^[399] These simulations allow for robust stress testing of investment strategies and risk models against a wide range of potential economic scenarios, enhancing financial stability and improving data driven decision making.

Finally, the future of AI in fintech will be shaped by the symbiotic evolution of technology and regulation. As AI's role expands, regulatory bodies are developing more sophisticated frameworks to govern its use, ensuring fairness, accountability, and stability.^[400] This has spurred the growth of Regulatory Technology (RegTech) and Supervisory Technology (SupTech), where AI itself is used as a tool for monitoring compliance and overseeing financial markets.^[401] The integration of AI into these supervisory functions enables regulators to analyze vast datasets in real time, identify systemic risks, and enforce regulations more effectively.^[402] This collaborative and technology enabled approach to governance will be essential for fostering responsible innovation while safeguarding the integrity of the global financial system.

Acknowledgments

The authors would like to acknowledge the support from EPSRC Grant Number EP/R00661X/1 & EP/P02470X/1.

Data Availability

Data sharing is not applicable to this research as no data were generated or analysed.

Ethical Approval

This article does not contain any studies with human participants performed by any of the authors.

Informed Consent

This article does not contain any studies with human participants performed by any of the authors.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supporting Information

Not applicable.

CRedit Statement

Zhizhou Zhang: Writing - Original draft, Writing – Review and editing, Visualisation, Validation, Methodology, Investigation, Conceptualisation, Funding acquisition, Project administration. **Meiqi Lu:** Writing - Review and editing. All authors have read and agreed to the published version of the manuscript.

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